

ASSESSMENT TOOL – 1

Course Code	CSA0702	Course Title	Computer Networks
CO Assessed	CO2	Assessment	Analytical Problem Solving
Weightage		Total Marks	

Student Name	Cindy R	Reg. No.	192412571
Section / Batch	ECE-2024	Date of Submission	29/7/26
Faculty Incharge	Dr R S Selvan	Assessment	Individual Submission

Total Marks	20	Obtained Marks	20
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Code of Conduct:

I Cindy R/192412571 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

University Network Solving

Given:

Network: 192.168.20.0/24

Lab A $\rightarrow$ /26

Lab B $\rightarrow$ /27

Lab C $\rightarrow$ /28

LAB A (/26)

Subnet Mask: 255.255.255.192

Network Address: 192.168.20.0

Host Range: 192.168.20.1 - 192.168.20.62

Broadcast Address: 192.168.20.63

Usable Hosts:

$$2^6 - 2 = 62$$

LAB B (/27)

Subnet Mask: 255.255.255.224

Network Address: 192.168.20.64

Hosted Range: 192.168.20.65 - 192.168.20.94

Broadcast Address: 192.168.20.95

Usable Host:

$$2^5 - 2 = 30$$

LAB C (/28)

Subnet Mask: 255.255.255.240

Network Address: 192.168.20.96

Host Range: 192.168.20.97 - 192.168.20.110

Broadcast Address: 192.168.20.111

Usable Hosts:

$$2^4 - 2 = 14$$

2. Analyze the Given systems

Given:

System 1

IP Addresses: 192.168.1.130

Mask: 255.255.255.192 (/26)

System 2

IP Address: 192.168.1.190

Mask: 255.255.255.192 (/26)

Step 1: Subnet Range

A /26 subnet has a block size of 64

subnets:

192.168.1.0-63

192.168.1.64-127

192.168.1.128-191

192.168.1.192-255

Both IP addresses fall within: 192.168.1.128/26.

Network Information:

Network address: 192.168.1.128

Broadcast address: 192.168.1.191

Range: 192.168.1.129 - 192.168.1.190

Verification:

System1 - 192.168.1.180 - Valid ✓

System2 - 192.168.1.190 - Valid ✓

Both belong to the same subnet

Usable Host

$$2^6 - 2 = 62$$

3. Organization Network Allocation

Given:

Network: 172.16.0.0/16

Departments: HR → /25

Sales → /26

IT → /27

Admin → /28

HR (/25)

Network: 172.16.0.0

Host Range: 172.16.0.1 → 172.16.0.126

Broadcast: 172.16.0.127

Hosts: 126

Sales (/26)

Network: 172.16.0.128

Host Range: 172.16.0.129 - 172.16.0.190

Broadcast: 172.16.0.191

Hosts : 62

IT (/27)

Network : 172.16.0.192

Host Range : 172.16.0.193 - 172.16.0.222

Broadcast : 172.16.0.223

Hosts : 30

Admin (/28)

Network : 172.16.0.224

Host Range : 172.16.0.225 - 172.16.0.238

Broadcast : 172.16.0.239

Hosts : 14

Remaining Unused Range

172.16.0.240 - 172.16.0.255

4. Subnet Identification:

Given:

Network : 172.16.10.0/24

Subnet Mask : /27

Given IP : 172.16.10.78

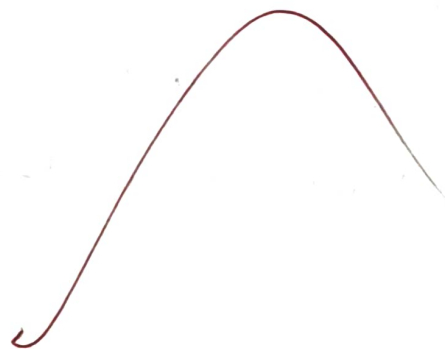
Step 1: No. of Subnets

Original Network : /24

New Network : /27

Borrowed Bits : 3

No. of subnets : $2^3 = 8$, Total subnets = 8.



p2: Host per subnet

Remaining Host Bits : 5

$$2^5 - 2 = 30$$

Usable Hosts = 30

Step 3: subnet of 172.16.10.78

Block size : 32

Subnets:

* 0 - 31

* 32 - 63

* 64 - 95

* 96 - 127

* 128 - 159

* 160 - 191

* 192 - 223

* 224 - 255

172.16.10.78 belongs to 172.16.10.64/27

Network Address : 172.16.10.64

Broadcast Address : 172.16.10.95

Host Range : 172.16.10.65 - 172.16.10.94

Does 172.16.10.95 belong to the same subnet?

Yes. However, 172.16.10.95 is the broadcast address, so it cannot be assigned to a host.

5. IPv6 Addressing

Given Address

2001:0dB8:0000:0000:0000:ff00:0042:8329

Compressed form: 2001:db8::ff00:42:8329

Expanded form:

2001:0db8:0000:0000:0000:ff00:0042:8329

Network prefix (164)

2001:db8:0000:0000::/64

Host bits:

$$128 - 64 = 64$$

$$\text{Total Host Addresses} = 2^{64} = 18,446,744,073,709,551,616$$

~~Q22-1~~